DIY Kick Stompbox

A kick drum for your foot. Play guitar and sing, and stomp on this pedal to add the kick. It plugs into your mixer or PA with a normal guitar cable, runs on a 9 V battery, and costs a few euros in parts.



The case. The violin-style f-holes and the hexagon can be printed in a second colour.

What it does

- **Stomp on the hexagon** and it plays a punchy kick drum.
- **Two sounds and one knob**: an acoustic kick and an 808-style kick. The knob blends them: fully left only the acoustic kick, fully right only the 808, in between a mix of the two.
- **Use your own sounds**: turn any WAV or MP3 into a sound for the pedal with the converter included in the files. With a single sound, the knob becomes its volume.
- **It ignores your foot resting on it**: lifting the foot does not play an extra kick.
- **Line output** on a 6.35 mm (1/4") jack: plug it into any mixer, audio interface or PA.
- **9 V battery** with an ON/OFF switch and a **battery door** underneath: no screws to change it.
- **LED** on top: on when the pedal is on, it blinks when the battery is getting low.
- **Sensitivity from 1 to 10**, set with one number in the program, and **built-in tests** to check the pedal before you close it.

How it works

Under the hexagon there is a piezo disc, the same kind of part used in buzzers and drum triggers. When you stomp, it sends a small electrical signal to an Arduino Nano, a tiny programmable board. The Arduino plays the kick sound, and a few resistors and capacitors bring it to the right level for your mixer.

Before you start. You will need to solder about 25 joints. Nothing in this project is dangerous: it runs on 9 V.

1. What you need

Electronics

Part	Qty	Notes
Arduino Nano	1	Original or clone, any USB connector.
Piezo disc, 27 mm, with wires	1	Usually sold in packs of 10-20.
6.35 mm (1/4") mono jack socket, panel mount	1	Mono (2 contacts).
10 kΩ potentiometer, linear, marked B10K + knob	1	Must be marked **B**, not A.
LED, 5 mm + LED holder	1	Any colour. For the holder, print any 5 mm LED holder for an 8 mm panel hole: there are many free ones on MakerWorld.
Diode 1N914 or 1N4148	2	Small glass diodes with a black band.
Resistor 1 MΩ	1	
Resistor 100 kΩ	2	One for the piezo, one for the battery check.
Resistor 47 kΩ	1	Battery check.
Resistor 2.2 kΩ	1	2 kΩ is fine too.
Resistor 1 kΩ	3	Two for the sound output, one for the LED.
Ceramic capacitor 47 nF (marked 473)	1	
Ceramic capacitor 100 nF (marked 104)	1	For the battery check.
Electrolytic capacitor 10 µF, 16 V or more	1	Has a + and a - leg: the - leg is marked with a stripe.

Power

Part	Qty	Notes
9 V battery	1	Lasts about 15 hours of playing. It goes in the compartment under the pedal, behind the battery door.
9 V battery clip	1	
Small slide switch	1	Three pins: only two are used.

Other parts

Part	Qty	Notes
M3 x 10 mm screws	4	To close the pedal (10 mm not counting the head).
Electrical tape	-	A small piece holds the Arduino in its seat. Use it also to insulate any bare joint.
Heat-shrink tubing	-	To cover the soldered joints so they cannot touch each other.
Thin wire	-	A few colours help.
Super glue	-	To glue the piezo disc.
Anti-slip rubber mat or rubber feet (optional)	-	Under the pedal so it does not slide. Leave the battery door free.

Tools

- Soldering iron and solder
- Multimeter
- Wire stripper and small cutters
- Computer with the free Arduino IDE, and a USB cable for the Arduino
- 3D printer

2. Printing

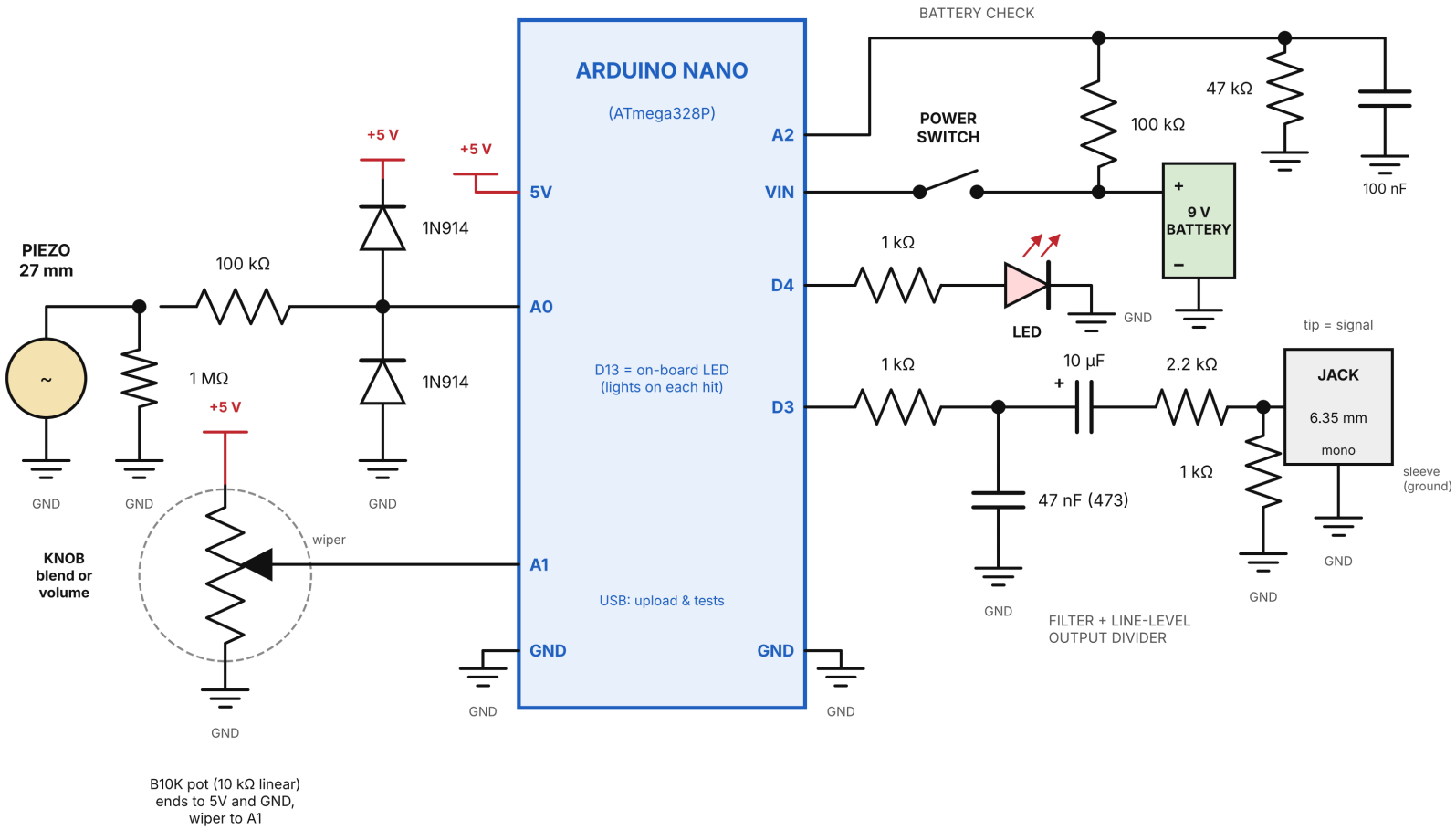
The pedal is printed in three parts: the **case**, the **bottom plate** and the **battery door**, plus a small LED holder (see below). The easiest way is to use the print profile on MakerWorld: everything is already set up on two plates.

- **Plate TOP - case**: printed standing up, with the back panel (jack, knob and switch holes) on the build plate, with supports where the slicer adds them.
- **Plate BOTTOM - bottom plate and battery door**: both printed flat, the battery door with its outside face on the build plate. The spring of the door lies flat too, so it bends along the printed lines and does not snap.
- **Colours**: the f-holes and the hexagon are separate parts set to the second filament (AMS). Change the colours in Bambu Studio or Bambu Handy before printing. In one colour they still show: the f-holes are engraved and the hexagon is raised.
- **Settings**: PLA, 0.2 mm layers, 4 walls, 25 % infill. Everything else as in the profile.

Tip. Before closing the pedal for good, try the battery door: it must click in, and open when you push its edge in the half-moon notch. The door has some play in its frame on purpose: it is normal that it moves a little.

DIY KICK STOMPBOX - wiring diagram

Arduino Nano + piezo + knob + power / battery LED + line output on a 6.35 mm (1/4") jack



HOW TO READ THIS DIAGRAM

- The three-line symbol is GND (ground): all these points are connected together.
- "+5 V" in red: all these points are connected to the Nano 5V pin.
- A black dot joins wires. Wires that cross without a dot are NOT connected.
- 1N914 diodes: the band (cathode) goes towards +5 V on the top one, towards the A0 wire on the bottom one.
- 10 µF electrolytic: + (long leg) towards the Arduino side, – (stripe) towards the jack.
- The battery goes to VIN, never to the 5V pin. The battery check is taken AFTER the power switch.
- LED: long leg (+) towards the 1 kΩ resistor, short leg (–, flat side) to GND.

3. Wiring

No circuit board is needed: solder the wires and the small parts directly to the pins of the Arduino, following the diagram on the previous page. Cover every joint with heat-shrink tubing. The name of each pin is printed on the Arduino board (A0, A1, A2, D3, D4, 5V, GND, VIN).

Part of the circuit	How to connect it
Piezo (pin A0)	Piezo red wire → 1 MΩ resistor to GND, and 100 kΩ resistor to A0. At A0: one diode to 5V (black band towards 5V) and one diode to GND (black band towards A0). Piezo black wire → GND.
Knob (pin A1)	Middle pin of the knob → A1. The two outer pins → 5V and GND. If the knob works the wrong way round, swap the two outer wires.
Sound output (pin D3)	D3 → 1 kΩ → node. Node → 47 nF capacitor to GND. Node → 10 µF capacitor (+ leg towards the node) → 2.2 kΩ → jack tip. 1 kΩ between jack tip and GND. Jack sleeve → GND.
LED (pin D4)	D4 → 1 kΩ → LED long leg (+). LED short leg (−, flat side of the LED) → GND.
Power	Battery red wire → switch → VIN. Battery black wire → GND.
Battery check (pin A2)	From the switch contact that goes to VIN (after the switch): 100 kΩ → A2. 47 kΩ between A2 and GND, and 100 nF between A2 and GND.

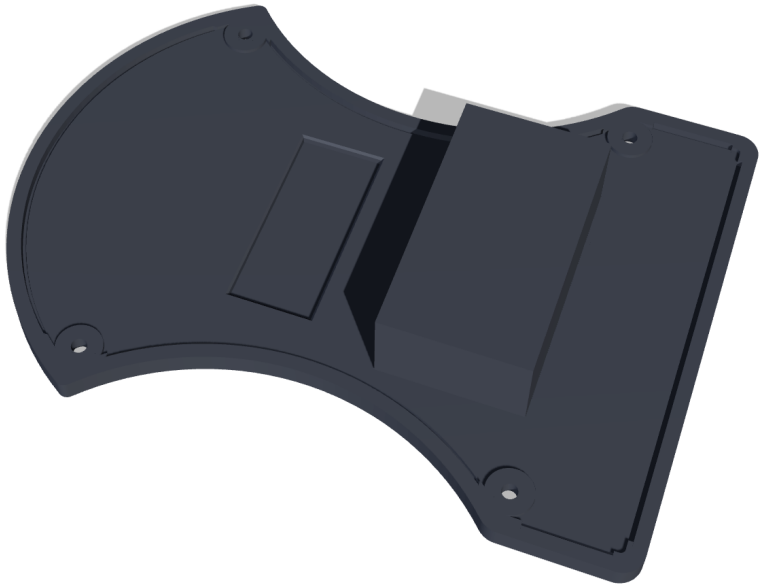
Finding the jack contacts

- **Sleeve**: the contact connected to the metal thread of the jack (check with the multimeter).
- **Tip**: plug in a cable and find the contact that connects to the very tip of the plug.
- If there is a third contact, leave it unconnected.

Soldering tips

- Do not heat the knob and the switch for more than 2-3 seconds per pin: their plastic melts and they stop working properly.
- The Arduino sits in the bottom plate and everything else is in the case: leave the wires long enough to open the pedal and lay the bottom plate next to it without pulling on the joints.
- Run the wires from the back panel to the Arduino along the sides of the battery compartment, where there is more room.

4. Assembly



Inside the bottom plate: the battery compartment and, next to it, the seat for the Arduino.

1. **Piezo.** Glue the disc under the top of the case, right under the hexagon, with a small drop of super glue. The metal side goes against the plastic. Press for 20 seconds. Do not use foam tape or a thick layer of glue: they soften the hit. Optional: a dot of glue on the wires 1-2 cm from the disc, so they do not pull on it.
2. **Jack, knob and switch** go in the back panel. The slot for the slide switch is tight on purpose, so the switch does not move when you use it: push it in firmly until it is fully seated.
3. **LED.** Snap the LED holder into the hole on top of the case, then push the LED into the holder from the inside. The holder is not in the print profile, but it is easy to find one to print: search MakerWorld for "LED holder 5 mm" and pick one for an **8 mm** hole (the hole in the case is 8.25 mm).
4. **Arduino.** Put it in its seat in the bottom plate and hold it down with a piece of **electrical tape**: the pedal stands on the floor, so the Arduino rests in its seat and the tape just stops it from moving. Check that no bare leg or joint can touch another one.
5. **Battery clip.** Pass the clip through the window at the end of the battery compartment.
6. **Load the program and run the tests** (sections 5-7) before closing the pedal.
7. **Close** it: the edge of the case fits into the groove of the bottom plate. Fix it with the four M3 x 10 mm screws, without overtightening them.
8. **Battery.** Put it in the compartment from underneath, with the clip on the side of the window, and close the battery door (below).
9. **Optional:** glue an anti-slip mat or rubber feet under the bottom plate, leaving the battery door free.

The battery door



Under the pedal: the battery door. Its far edge is a spring; the hook on it goes up through the bottom plate and catches inside.

- **To close it**: put the two tabs on the hinge side into their slots first, then press the other end until it clicks. Don't worry if the closed door moves a little: the play is there so the hook always reaches its place.
- **To open it**: put a fingertip in the half-moon notch and push the edge of the door towards its middle: the edge is a spring: about 2 mm is enough for the hook to let go, and the door comes out.
- To take the battery out, hold the clip itself, not its wires: the clip holds tight on the battery.
- **Tip**: put a small piece of foam or folded cloth in the battery compartment, next to the battery, so it does not rattle when you stomp.

5. Loading the program

1. Install the **Arduino IDE** from arduino.cc (free).
2. Unzip the firmware file. Keep the folder name **stompbox_kick** and keep all the files inside it.
3. Double-click `stompbox_kick.ino` to open it in the Arduino IDE.
4. In the **Tools** menu choose **Board → Arduino Nano**, and **Processor → ATmega328P**.
5. Connect the USB cable, choose the **Port** in the Tools menu and press **Upload** (the arrow button).
6. If the upload fails, choose **Processor → ATmega328P (Old Bootloader)** and try again: many Arduino clones need it.

6. Sounds

The sounds are files in the `stompbox_kick` folder, next to the program. The program looks at which ones are there and sets the knob by itself:

Files in the folder	What the knob does
`sample_a.h` + `sample_b.h` (as supplied: acoustic kick + 808)	Blends the two sounds: fully left only sound A, fully right only sound B.
Only one of the two	It is the volume of that sound (fully left = silent).

Using your own sounds

1. Open `sample_converter.html` (in the firmware zip) with a double-click: it opens in your browser, on Windows or Mac, and works without internet. Nothing is uploaded.
2. Drop in an audio file with a single hit (WAV or MP3), choose **Sound A** or **Sound B**, listen to it and save it.
3. Put the saved file in the `stompbox_kick` folder, replacing the old one.
4. If the program is open in the Arduino IDE, close it and open it again, then upload it.

- Each sound can be up to 0.68 s long: plenty for a kick.
- The converter does the rest by itself: it removes the silence at the start, sets the level and fades out the tail.
- The two supplied sounds are also in the `spare_sounds` folder, if you want them back.

7. Testing before you close

With the USB cable still connected, open **Tools → Serial Monitor** and set it to **115200 baud** (bottom right). The first lines say which sounds the pedal is using. Type a number and press Enter:

Type	Test	What should happen
3	Jack	A beep for 2 seconds on your mixer or speaker. No beep: check the sound output parts, the jack, the cable and the mixer.
1	Piezo	A line of numbers ten times a second. With the pedal still, the number stays low (under 20-30). When you stomp it goes much higher and says "above threshold".
2	Knob	Turn the knob from end to end: the number goes from 0 to 1023, smoothly.
4 / 5	Sounds	Plays sound A / sound B once (with a single sound, 4 plays it).
6	Battery	Switch the pedal ON: it shows the battery voltage. A new battery reads 9.0-9.6 V.
0	Stop	Stops the tests.

When everything works, the Serial Monitor shows a line "kick" every time you stomp.

8. Sensitivity

If the pedal plays by itself, or light stomps do not play, change the sensitivity. It is the first setting in the program, inside a big box of # signs:

```
#define SENSITIVITY 5
```

- Write a number from **1** (only hard stomps play) to **10** (light taps play too). 5 is the starting point.
- 5 is tuned for playing with the **heel**. If you play with the **tip of your foot**, raise it a little (6 or 7).
- It plays by itself → **lower** the number.
- Light stomps do not play → **raise** the number.
- Then upload the program again (section 5).

9. Playing

- Use a normal guitar cable into a **line input** of your mixer, audio interface or PA.
- For a microphone input (XLR) use a DI box.
- A kick drum is a very low sound: small speakers will only play the click. Use a PA or a speaker with good bass.
- Switch the pedal OFF when you are not playing, to save the battery.

The LED and the battery

LED	Meaning
On	The pedal is on and the battery is OK (or the pedal is powered from the USB cable).
Slow blink	The battery is low (under 7 V): change it soon.
Fast blink	The battery is almost empty (under 6.5 V): change it now.

A 9 V alkaline battery lasts about **15 hours** with the pedal switched on. To change it, open the battery door under the pedal: no screws needed.

10. Problems and solutions

Problem	What to check
It plays by itself all the time	A loose ground (GND) wire or a missing 1 MΩ resistor at the piezo. Run test 1: with the pedal still, the number must stay low. If it does, lower SENSITIVITY.
It works only if you move the wires	A bad solder joint. Resolder it.
The knob jumps or does nothing	Run test 2. Check the three knob wires. If it still jumps, the knob was probably overheated while soldering: replace it.
No sound	Run test 3. Check the 10 µF capacitor (+ and - legs), the jack contacts, the cable and the mixer input.
Light stomps do not play	Raise SENSITIVITY (section 8).
The LED does not light	It is probably the wrong way round: the long leg goes towards the 1 kΩ resistor.
The LED blinks with a new battery	Run test 6. If it reads much less than 9 V, check the 100 kΩ and 47 kΩ resistors of the battery check.
The Arduino IDE says "No sound in the folder"	Put sample_a.h and/or sample_b.h in the stompbox_kick folder, then close and reopen the program.
My new sound does not play	The file must be in the stompbox_kick folder and named sample_a.h or sample_b.h. Close and reopen the program before uploading.
Upload fails	Choose ATmega328P (Old Bootloader) and check the port.
Sound too quiet or distorted	Use a line input and adjust the gain on the mixer. For a microphone input use a DI box.

Licence and credits

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- **3D model**: Standard Digital File License - personal use only; no sharing of the files, no remixes, no selling of the files or of printed parts.
- **Program, sounds, sample converter, wiring diagram and this guide**: Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 (CC BY-NC-ND 4.0). You may change the settings and the sounds for your own pedal, but you may not publish modified versions or use them commercially.
- **Kick sounds**: the acoustic kick and the 808 are original sounds made for this pedal.

Designed by 3Dvarius. Program written with the help of Claude (Anthropic).